EX NO:2 Chebyshev IIR Filter

2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

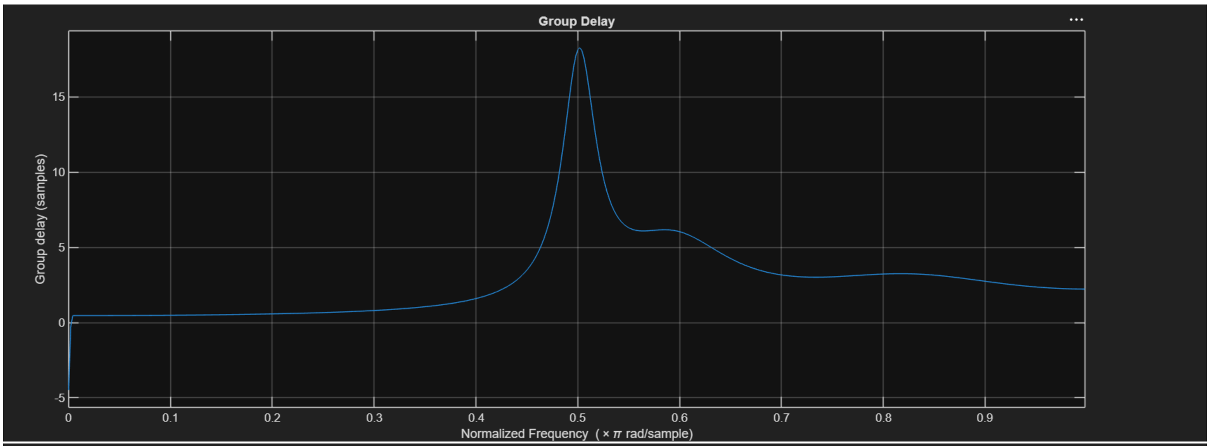
PROGRAM

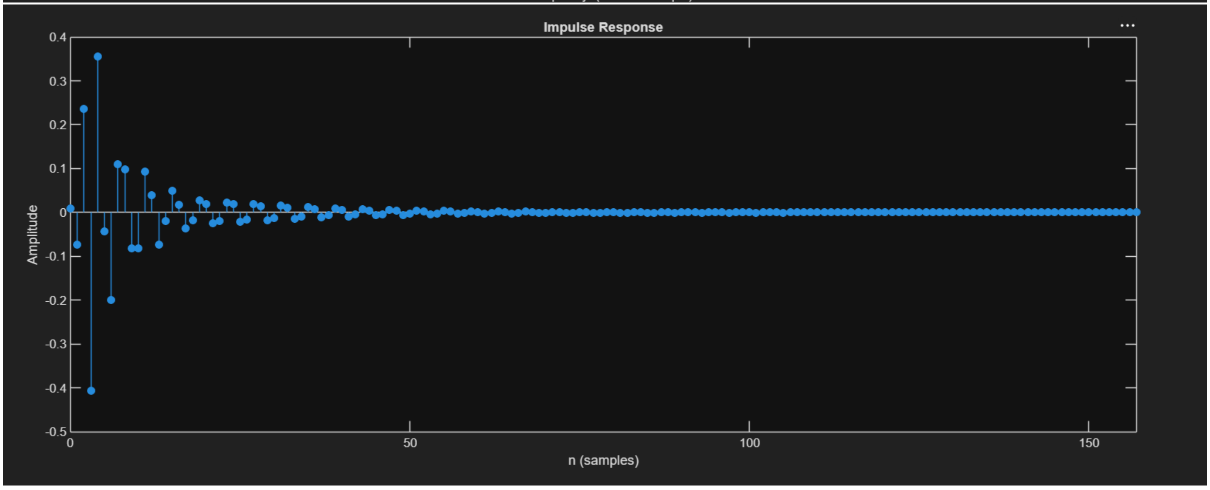
```
clc;
clear;
close all;
% Specifications
Rp = 1;      % Passband Ripple (dB)
Rs = 40;     % Stopband Attenuation (dB)
Wp = 0.5;    % Passband Edge
Ws = 0.35;   % Stopband Edge
% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');
% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');
% Impulse Response
figure;
impz(b,a);
title('Impulse Response');
% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');
% Pole-Zero Plot
```

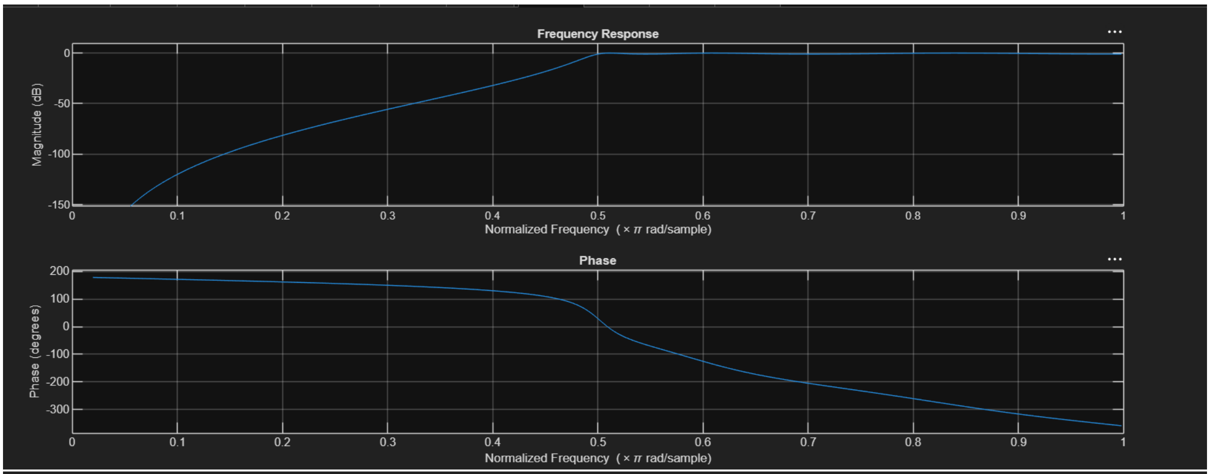
```
figure;
zplane(b,a);
title('Pole-Zero Plot');
fprintf('Filter Order = %d\n',n);
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

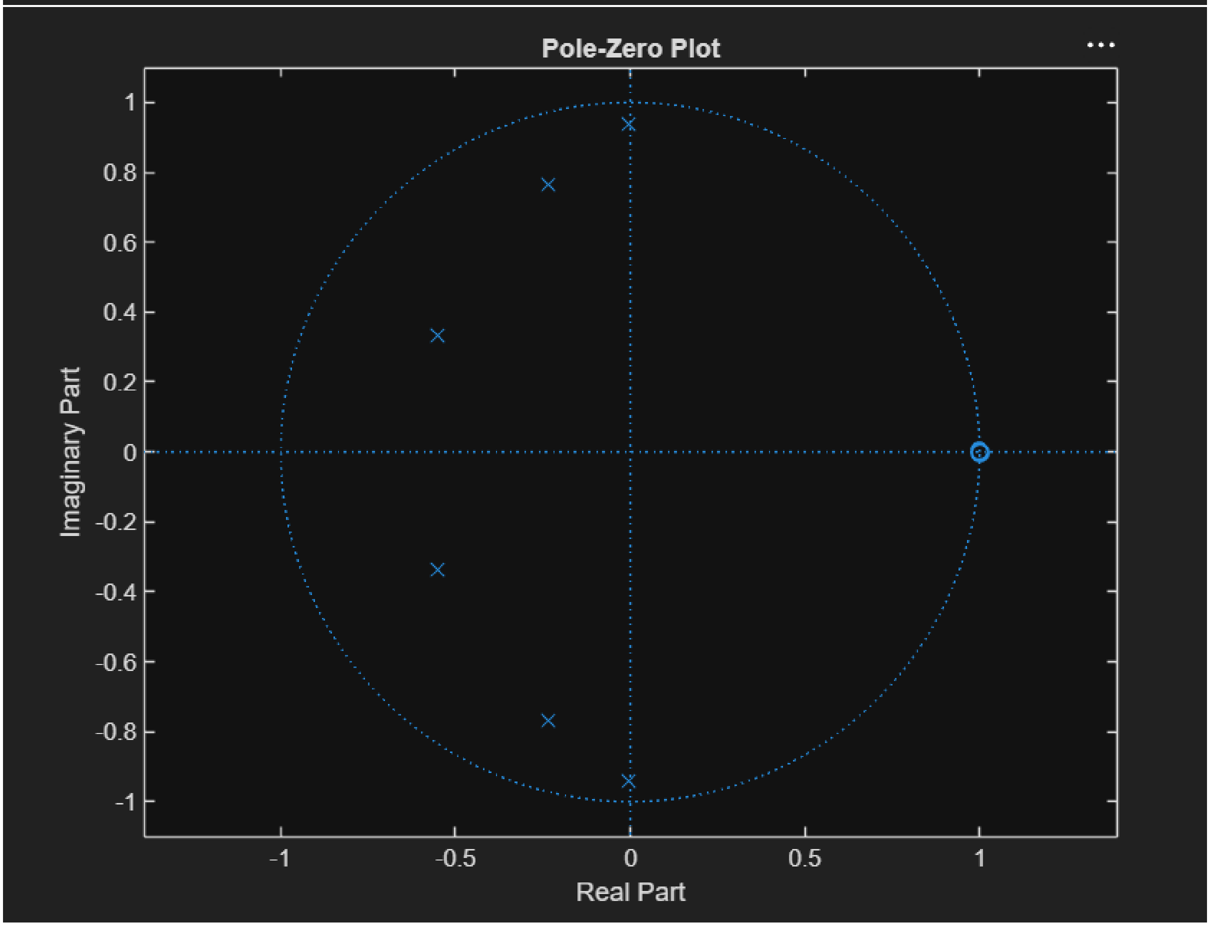
OUTPUT

Filter Order = 6
Cutoff Frequency = 0.5000






Frequency Response
Magnitude (dB)
0
-50
-100
-150
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
200
100
0
-100
-200
-300
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)


Pole-Zero Plot
Imaginary Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1
Real Part
3